Part 3: MARIE Programming (18 marks total)

In this task you will develop a MARIE application that performs arithmetic calculation based on user input, display the result in the output panel, as well as the MARIE bitmap panel. We will break it down into steps for you.

Each task requires you to write **code, including appropriate labels** and **documentation**. On Moodle, you will find a **template** for Part 3. **Your submission MUST BE based on the given template only**, i.e., you must add implementations of your own subroutines into the template under relevant sections. The template already contains the **main program** that calls the subroutines.

Your code must contain readable comments and meaningful labels for your tutor / marker to understand the logic flow of your program (e.g., the **purpose** of a subroutine, `Jump` / `SkipCond` instruction etc.).

In-class demonstration (Part 4): You will be required to join a demonstration to show your code (and successfully run or execute your Part 3 tasks to be considered for grading) to your tutor during your applied session in Week 8 [Australian Cohort] (Malaysian Cohort TBC for the time). **Failure to demonstrate** (successfully run or execute your Part 3 tasks**) will lead to zero marks being awarded for the entire Part 3, regardless of your submission in Moodle**. In addition, during the demonstration (Part 4), you will also need to answer further questions about your submitted Part 3 tasks (see below for details) and MARIE and OS fundamentals.

Code similarity: We use tools such as MOSS and Turnitin to check for collaboration and copying between students. If you copy parts of your code from other students, or you let them copy parts of your code, this will result in a report to the Academic Integrity team. As a result, **you may receive a penalty such as 0 marks for the entire assignment, 0 marks for the whole unit, or in severe cases (such as contract cheating), suspension or expulsion from Monash University**.

Tasks Background

Arithmetic calculation

Arithmetic calculation is a fundamental computing task in which a program interprets numerical input, applies a selected operator, and produces a correct result. In this assignment, you will build a MARIE application that reads a user-entered arithmetic expression (in the "Inputs" panel as Unicode (UTF-16BE)) in a predefined format, that includes two positive integers separated by one operator from +, -, * or /, followed by "=?" (without any space in between). For example, the following inputs (as Unicode (UTF-16BE)):

```
123+456=?
159-23=?
59-83=?
```